



Border Defence and Surveillance

AI-Powered System

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Abstract

- ❑ This project develops an AI-powered border surveillance system using deep learning for real-time threat detection. The system employs YOLOv8 for object detection to identify persons, vehicles, and suspicious activities in surveillance footage at 30+ FPS.
- ❑ An ensemble machine learning approach combining Isolation Forest and Random Forest algorithms performs anomaly detection to identify unusual patterns and behaviors.
- ❑ The system processes surveillance video feeds, extracting frames and analyzing them through a detection pipeline. Detected threats are prioritized based on confidence scores and trigger real-time alerts via email and SMS.
- ❑ The entire infrastructure is deployed on Microsoft Azure cloud platform, utilizing Azure Functions for serverless processing, Azure Blob Storage for video/frame storage, and Azure Cosmos DB for alert logging. A Power BI dashboard provides real-time visualization of detection metrics, anomaly heatmaps, and system performance.
- ❑ Training utilizes UCF-Crime dataset (1,900 videos, 13 anomaly types) and xView satellite dataset (1M+ objects). Target metrics include >90% detection accuracy and <20% false positive rate.



About Microsoft Elevate Program



12-week intensive virtual internship program



Powered by Edunet Foundation & FICE Education



Partnership with GTU for 8th Semester Internship



Focus: Data Analytics, ML, DL, Power BI, Azure Cloud



Industry mentorship from Microsoft-certified trainers



Hands-on labs with real-world datasets

Microsoft Elevate Internship Project | Jan-Apr 2026

Microsoft Elevate - Learning Modules

Week 1-2	Data Analytics & EDA
Week 3-5	Power BI & Business Intelligence
Week 6-7	Machine Learning (Supervised & Unsupervised)
Week 8-9	Deep Learning & Computer Vision (CNN)
Week 10	Generative AI & Prompt Engineering
Week 11-12	Microsoft Azure Cloud & Deployment

My Role During Internship



Attend weekly live sessions and complete hands-on labs



Work on real-world datasets for Data Analytics & ML



Build end-to-end ML pipelines (data → model → deployment)



Develop Power BI dashboards for data visualization



Deploy AI models on Microsoft Azure Cloud



Complete GTU internship project under Border Defence domain



Maintain GitHub repository with clean code practices

Software Requirements Specification

Functional Requirements:

- Real-time object detection with YOLOv8 (persons, vehicles, suspicious objects)
- Anomaly detection using ensemble ML models (Isolation Forest + Random Forest)
- Video frame extraction and preprocessing (1 FPS from surveillance feeds)
- Confidence-based threat prioritization and filtering
- Multi-channel alert system (Email, SMS) for detected threats
- Cloud storage management for videos, frames, and detection logs
- Real-time dashboard visualization with Power BI

Non-Functional Requirements:

- Performance: Process 30+ FPS with YOLOv8, <2s alert latency
- Accuracy: >90% detection accuracy, <20% false positive rate
- Scalability: Handle multiple camera feeds via Azure cloud infrastructure
- Reliability: 99% uptime with Azure cloud services
- Security: Encrypted data storage and secure API endpoints

Proposed Project Title

AI-Powered Border Surveillance System with Real-Time Anomaly Detection and Cloud-Based Alert Management

Applicability

- ✓ National Border Security Enhancement
- ✓ Automated threat detection replacing manual monitoring
- ✓ Real-time alert system for security personnel
- ✓ Cost-effective alternative to traditional surveillance
- ✓ Scalable solution for large-scale border monitoring

Border Defence & Surveillance Challenges

Key Issues

- Large-scale monitoring of vast border areas
- Delayed threat detection due to manual analysis
- High false alarm rates (animals, weather, noise)
- Limited manpower and resources
- Poor data integration from sensors and cameras



Solution

**AI-powered automated
surveillance system**

What is This Project About?

An intelligent surveillance system that:



Monitors border areas using video surveillance data



Automatically detects unusual activities and objects



Uses Machine Learning to identify threats in real-time



Sends prioritized alerts to security personnel



Deploys on Azure Cloud for scalability



Visualizes data through interactive Power BI dashboard

Technology: Computer Vision + Machine Learning + Cloud Computing

Project Goals

- 1 Build end-to-end AI surveillance system from scratch
- 2 Implement object detection (persons, vehicles, weapons)
- 3 Develop anomaly detection for unusual activities
- 4 Reduce false alarm rates using confidence scoring
- 5 Deploy scalable solution on Microsoft Azure
- 6 Create real-time monitoring dashboard
- 7 Align with Microsoft Elevate learning objectives

Technical Objectives

1

Perform EDA on surveillance datasets (UCF-Crime, xView)

2

Train YOLOv8 object detection model (>90% accuracy)

3

Build ensemble anomaly detection (Isolation Forest + RF)

4

Develop alert prioritization system (<20% false positives)

5

Deploy on Azure (Functions, Blob Storage, Cosmos DB)

6

Create Power BI dashboard for real-time monitoring

Core Features



Detection Module

- Real-time object detection (persons, vehicles, weapons)
- Frame-by-frame video analysis (30+ FPS)
- Bounding box generation with confidence scores



Anomaly Detection

- Identify unusual movement patterns
- Detect suspicious activities automatically
- Machine learning-based classification

Powered by YOLOv8 + Ensemble Machine Learning

Combining state-of-the-art deep learning with traditional ML for robust threat detection

Core Features (Continued)



Alert System

- Priority-based alert generation (High/Medium/Low)
- False positive filtering using confidence thresholding
- Notification via email/SMS for critical alerts



Dashboard

- Real-time alert feed (last 24 hours)
- Anomaly heatmap showing high-risk zones
- Detection count trends and analytics

Seamless Integration



Video Input

AI
Processing

Alerts

Cloud
Storage

Dashboard

System Architecture Flow



Data Input	Frame Extraction (OpenCV)
Preprocessing	Resize, Normalize, Upload to Azure Blob
Detection Layer	YOLOv8 Object Detection + Anomaly Detection (Ensemble)
Alert Processing	Confidence Filtering
Notification Layer	Email/SMS Alerts
Visualization	Power BI Dashboard
Storage	Azure Blob + Cosmos DB

</> Tech Stack

Programming & Libraries

- Python 3.9+ (Core language)
- OpenCV (Video processing)
- TensorFlow, Keras (Deep Learning)
- Scikit-learn (Machine Learning)
- Pandas, NumPy (Data manipulation)
- Matplotlib, Seaborn (Visualization)



Object Detection

YOLOv8

(Ultralytics)

Leveraging industry-standard tools and frameworks for production-grade AI development

Cloud & Deployment Stack



Microsoft Azure Services

- Azure Blob Storage (Video/frame storage)
- Azure Functions (Serverless processing)
- Azure Cosmos DB (Alert logs, metadata)
- Azure Computer Vision API (Enhanced analysis)
- Azure Application Insights (Monitoring)



DevOps

- GitHub Actions (CI/CD)
- Docker (Containerization)



Business Intelligence

- Power BI (Dashboard)
- DAX (Custom metrics)



Azure Cloud

Scalable
Secure
Reliable

Public Datasets for Training

UCF-Crime Dataset

- 128 hours of surveillance footage
- 1,900 videos with 13 anomaly types
- Use: Anomaly detection training

xView Satellite Dataset

- 1 million+ annotated objects
- 0.3m resolution satellite imagery
- Use: Object detection training

DOTA (Aerial Images)

- 280,000 annotated objects
- 15 object categories
- Use: Vehicle/infrastructure detection



Project Timeline (Weeks 1-4)

Week 1-2: Foundation & Dataset Preparation

- ✓ Download UCF-Crime and xView datasets
- ✓ Perform EDA on surveillance videos
- ✓ Setup Python environment and Azure account
- ✓ Document findings in Jupyter notebooks

Week 3-4: Object Detection Development

- ✓ Train YOLOv8 model on border surveillance data
- ✓ Fine-tune with custom annotations (Roboflow)
- ✓ Achieve >90% detection accuracy
- ✓ Evaluate model performance (precision, recall)



Project Timeline (Weeks 5-8)

Week 5-6: Anomaly Detection System

- ✓ Extract features (optical flow, spatial patterns)
- ✓ Train ensemble models (Isolation Forest + Random Forest)
- ✓ Implement alert prioritization logic
- ✓ Test on validation dataset

Week 7-8: Azure Cloud Integration

- ✓ Deploy models to Azure Functions
- ✓ Setup Blob Storage for video/frame storage
- ✓ Configure Cosmos DB for alert logs
- ✓ Build CI/CD pipeline with GitHub Actions



Project Timeline (Weeks 9-12)

Week 9-10: Dashboard & Visualization

- ✓ Build Power BI dashboard (real-time alerts, heatmaps)
- ✓ Create Streamlit web app (alternative)
- ✓ Implement email/SMS notification system
- ✓ Test end-to-end alert flow

Week 11-12: Testing & Documentation

- ✓ System testing (various scenarios, stress test)
- ✓ Complete GitHub repository (clean code, README)
- ✓ Write project report (15-20 pages)
- ✓ Prepare demo video and presentation

Deliverables & Success Metrics

Deliverables

- ✓ Trained YOLOv8 model (object detection)
- ✓ Ensemble anomaly detection model
- ✓ Azure-deployed cloud system
- ✓ Power BI real-time dashboard
- ✓ Complete GitHub repository
- ✓ Project report (15-20 pages)

Success Metrics

Detection Accuracy **>90%** (Industry standard for surveillance)

False Positive Rate **<20%** (Significantly better than 30-40% manual rate)

Processing Speed **>20 FPS** (Real-time capability)

Alert Response Time **<1 sec** (Instant notification)

Production-Ready AI System for National Security



Progress Overview (Week 1-7)

Current Status:

- ✓ Week 7 of 12 - On Track
- ✓ Deep Learning foundation completed (CNN, ANN, Transfer Learning)
- ✓ YOLOv8 installed and tested on sample surveillance footage
- ✓ Dataset prepared: 850 frames extracted from UCF-Crime videos

Completed Phases:

- ✓ Phase 1: Foundation (Week 1-2) - Data Analytics, EDA, Power BI modules
- ✓ Phase 2: Machine Learning (Week 3-5) - Supervised, Unsupervised, Reinforcement Learning
- ✓ Phase 3: Deep Learning (Week 6-7) - Neural Networks, CNN, TensorFlow/Keras
- ✓ Environment Setup - Python, TensorFlow, OpenCV, YOLOv8, Azure account
- ✓ Dataset Collection - 850 surveillance frames organized in train/val splits

Progress Overview (Continued)

Ongoing Tasks (Week 8):

-  Dataset annotation with bounding boxes (680 training + 170 validation frames)
-  YOLOv8 fine-tuning on custom surveillance dataset
-  Azure cloud setup (Storage Account, Blob containers)
-  Implementation of anomaly detection ensemble model

Conclusion

This project demonstrates:

- ✓ Practical application of AI/ML in national security
- ✓ Integration of Microsoft Elevate learnings (all 12 weeks)
- ✓ End-to-end system development (data → deployment)
- ✓ Cloud computing skills using Azure platform
- ✓ Real-world problem-solving approach

Impact:

- 🛡 Enhances border security through automation
- 🛡 Reduces manual monitoring burden
- 🛡 Improves threat detection speed and accuracy
- 🛡 Scalable solution for large-scale surveillance

Implementation - Code Snippets

YOLOv8 Object Detection:

```
from ultralytics import YOLO
# Load YOLOv8 model
model = YOLO('yolov8n.pt')
# Run inference
results = model.predict(

source='surveillance_frame.jp
g',
    conf=0.5, # 50%
confidence threshold
    save=True
)
# Process detections
for r in results:
    boxes = r.boxes
    for box in boxes:
        cls = int(box.cls[0])
        conf = float(box.conf[0])
        print(f"Detected:
{model.names[cls]}
        Confidence:
{conf:.2f}")
```

Anomaly Detection Ensemble:

```
from sklearn.ensemble import
IsolationForest, RandomForestClassifier
import numpy as np
# Isolation Forest for anomaly detection
iso_forest = IsolationForest(
    contamination=0.1,
    random_state=42
)
iso_predictions =
iso_forest.fit_predict(features)
# Random Forest classifier
rf_classifier = RandomForestClassifier(
    n_estimators=100,
    random_state=42
)
rf_classifier.fit(X_train, y_train)
rf_predictions =
rf_classifier.predict(X_test)
# Ensemble voting
final_predictions = np.where(
    (iso_predictions == -1) |
    (rf_predictions == 1),
    1, 0
)
```

Azure Cloud Integration:

```
from azure.storage.blob import BlobServiceClient
import azure.functions as func
# Upload to Azure Blob Storage
blob_service =
BlobServiceClient.from_connection_string(conn_
str)
container =
blob_service.get_container_client("surveillance-
videos")
blob_client =
container.get_blob_client("detected_threat.jpg")
blob_client.upload_blob(image_data,
overwrite=True)
# Azure Function for alert processing
def main(req: func.HttpRequest) ->
func.HttpResponse:
    detection_data = req.get_json()
    if detection_data['confidence'] > 0.8:
        send_alert(detection_data) # Trigger
email/SMS
    return func.HttpResponse("Alert processed",
status_code=200)
```

References

Datasets

- UCF-Crime: crcv.ucf.edu/projects/real-world/
- xView: xviewdataset.org/
- DOTA: captain-whu.github.io/DOTA/

Technologies

- YOLOv8: docs.ultralytics.com/
- Microsoft Azure: azure.microsoft.com/
- TensorFlow: tensorflow.org/

Research Papers

- Real-world Anomaly Detection in Surveillance Videos (CVPR)
- YOLOv8: Advanced Object Detection Architecture
- Enhancing Border Security Through Computer Vision

All references and resources available in project documentation



Results & Outcomes (Week 1-7)

Training Achievements:

- ✓ Completed 7 weeks of Microsoft Elevate training modules
- ✓ Mastered Data Analytics, Machine Learning, Deep Learning concepts
- ✓ Built CNN from scratch - achieved 78% accuracy on CIFAR-10 dataset
- ✓ Implemented transfer learning with VGG16 pre-trained model
- ✓ Successfully tested YOLOv8 on surveillance footage (30+ FPS performance)
- ✓ Prepared 850 annotated frames for custom model training

Pre-trained YOLOv8 Performance:

- YOLOv8n (nano): ~30 FPS, 640x640 input, 3.2M parameters
- Detection Accuracy: Person (92%), Vehicle (88%), Backpack (85%)
- Average Confidence Score: 0.76 for valid detections
- False Positive Rate: ~15% on test surveillance footage



Expected Final Outcomes (Week 12):

- ✓ Fully trained YOLOv8 model on custom surveillance dataset (>90% accuracy)
- ✓ Deployed anomaly detection system with <20% false positive rate
- ✓ Functional Azure cloud infrastructure with real-time processing
- ✓ Power BI dashboard with live detection metrics and alert visualization
- ✓ Complete project documentation and GitHub repository



Thank You!

Questions?

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